Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2024-P2-Model)** **Date: 15-03-2026**

Time: 02.00Pm to 05.00Pm **GTA-18** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**D**	2	**D**	3	**C**	4	**C**	5	**BCD**	6	**ABC**
7	**BD**	8	**89**	9	**5**	10	**3**	11	**4**	12	**17**
13	**5**	14	**2.1-2.3**	15	**2.3-2.4**	16	**1.6-1.7**	17	**2**		

PHYSICS

18	**B**	19	**B**	20	**C**	21	**A**	22	**AC**	23	**ABCD**
24	**ABC**	25	**5**	26	**5**	27	**6**	28	**2**	29	**4**
30	**15**	31	**20**	32	**10**	33	**19**	34	**9**		

CHEMISTRY

35	**C**	36	**C**	37	**A**	38	**C**	39	**ACD**	40	**ABC**
41	**ACD**	42	**1**	43	**81**	44	**3**	45	**1**	46	**8**
47	**15**	48	**0.67**	49	**71.44**	50	**9**	51	**4**		

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SOLUTIONS

MATHEMATICS

1. Let a, b, c are the roots of equation

$a_0x^3+a_1x^2+a_2x+a_3=0$

Then, $s_1=5 \quad s_2=12, \quad s_3=25$

Now, $a_0s_1+a_1=0 \Rightarrow a_1=-5a_0$

$a_0s_2+a_1s_1+2a_2=0 \Rightarrow a_2=\frac{13}{2}a_0$

$a_0s_3+a_1s_2+a_2s_1+3a_3=0 \Rightarrow a_3=\frac{5}{6}a_0$

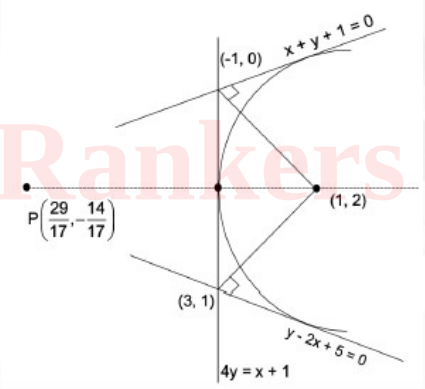
$x^3-5x^2+\frac{13}{2}x+\frac{5}{6}=0$

$6x^3-30x^2+39x+5=0 \qquad \Rightarrow 6x^4-30x^3+39x^2+5x=0$

$6s^4-30s_3+39s_2+5s_1=0 \Rightarrow s_4=\frac{257}{6}$

2. Normal through given point be the axis of parabola

$y-2=-4(x-1) \quad 4x+y=6$



3. $|z_2+iz_1|=|z_1|+|z_2| \Rightarrow z_2, iz_1 0$ are collinear

$z_3=\frac{z_2-iz_1}{1-i} \Rightarrow (1-i)z_3=z_2-iz_1 \Rightarrow i=\frac{z_3-z_2}{z_3-z_1} \Rightarrow \arg\left(\frac{z_2-z_2}{z_3-z_1}\right)=\frac{\pi}{2}$

and $|z_3-z_2|=|z_3-z_1| \qquad \Rightarrow AC=BC, AB^1=AC^2+BC^2$

$\Rightarrow AC=\frac{5}{\sqrt{2}}; \Delta\frac{1}{2}\times\frac{5}{\sqrt{2}}\times\frac{5}{\sqrt{2}}=\frac{25}{4} sq.units$

4. Clearly incentre of ΔDEF is orthocenter of ΔABC is i.e (α,α) which lies on hyperbola

$xy-x-y=3=0 \Rightarrow \alpha=3$

Ahain equation of DE is $x+y-\frac{26}{5}=0$

$\Rightarrow r=$ length of perpendicular from

$I(3,3)$ on $DE \Rightarrow r=\frac{2}{5}\sqrt{2}$ units

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5. $S_n = \sum_{k=1}^{\infty} k^n r^k$

$$\frac{d}{dr}(S_n) = \sum_{k=1}^{\infty} k^n . k r^{k-1} = \frac{1}{r}\sum_{k=1}^{\infty} k^{n-1} r^n$$

$$= r\left[\frac{(1-r)^2 + 2r(1-r)}{(1-r)^4}\right] = \frac{r}{(1-r)^2} + \frac{2r^2}{(1-r)^3} = \frac{1}{r}\left[S_0^2 + 2S_0^3\right]$$

6. $\angle B = \tan^{-1}\left(\frac{\frac{3}{4}+\frac{1}{2}}{1-\frac{3}{4}.\frac{1}{2}}\right) = \tan^{-1} 2$ $\qquad \angle C = \tan^{-1}\left(\frac{\frac{7}{24}+\frac{13}{9}}{1-\frac{7}{24}.\frac{13}{9}}\right) = \tan^{-1} 3$

$\angle A = \pi - \angle B - \angle C = \tan^{-1} 1$ $\qquad \therefore \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

7. $P(A_3) = \frac{{}^6C_3 \times \left(\frac{6!}{2!2!2!} + \frac{6!}{1!2!3!} \times 3! + \frac{6!}{1!1!4!} \times 3\right)}{6^6} = \frac{10800}{6^6}$

$$P(A_4) = \frac{{}^6C_4\left(\frac{6! \times 4!}{3!(1!)^3 3!} + \frac{6!}{(2!)^4 (1!)^2}\right)}{6^6} = \frac{23400}{6^6}$$

$$P(A_5) = \frac{{}^6C_5\left({}^5C_1 \times \frac{6!}{2!}\right)}{6^6} = \frac{10800}{6^6}$$

8. If x_1, x_2, x_3 *and* x_4 are the root of p then $\frac{p'(7)}{p(7)} = \sum \frac{1}{7-x_i} = 0$ (as 7 is not a root of p(x))

$$\Rightarrow \frac{1}{7-1} + \frac{1}{7-3} + \frac{1}{7-5} + \frac{1}{7-x_4} = 0 \Rightarrow x_4 = \frac{89}{11}$$

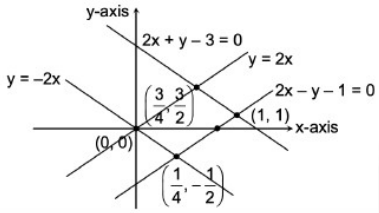
9. Let $S_n = \int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} f(x)dx$, then $S_n = \frac{1}{3}\int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} f(2x)dx = \frac{S_{n-1}}{6} = \ldots\ldots = \frac{S_0}{6^n}$

$$\int_0^1 f(x)dx = \lim_{n\to\infty}\left(\int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} f(x)dx + \ldots. + \int_{\frac{1}{2^{n-1}}}^{1} f(x)dx + \ldots. + \int_{\frac{1}{2}}^{1} f(x)\right) = \frac{S_0}{5} = 1 \Rightarrow S_0 = 5$$

10. lines with slope ± 2 and passing through (0, 0) and (1, 1) are $y = \pm 2x$ and $y - 1 = \pm 2(x-1)$ respectively

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By the Mean value theorem, no point of form (x, f(x)) lies outside this parallelogram. However, we given condition and are arbitrarily close to the sides of the parallelogram.
So, b-a = area of parallelogram = ¾



11. Give plane can be

$(2x+3y-z+1)+\lambda(x+y-2z+3)=0$

$\Rightarrow x(2+\lambda)+y(3+\lambda)-z(1+2\lambda)+1+3\lambda=0$

It is perpendicular to $3x-y-2z-4=0$

$\Rightarrow 3(2+\lambda)-(3+\lambda)+2(1+2\lambda)=0 \qquad \Rightarrow 6\lambda+5=0\Rightarrow\lambda=-\frac{5}{6}$

$\therefore$ Plane is $7x+13y+4z-9=0\therefore l=7 \quad m=4 \quad n=-9\ \frac{l-n}{m}=\frac{16}{4}=4.00$

12. $\left(\tan^{-1}y-x\right)dy=\left(1+y^2\right)dx \qquad \Rightarrow\frac{dx}{dy}+\frac{x}{1+y^2}=\frac{\tan^{-1}y}{1+y^2}$

$\therefore x.e^{\tan^{-1}y}=e^{\tan^{-1}y}\left(\tan^{-1}y-1\right)+c \qquad \Rightarrow x=\tan^{-1}y-1+ce^{-\tan^{-1}y}$

13. Point P lie on line $y-2=-\frac{1}{2}(x-1) \qquad 2y-4=-x+1 \qquad x+2y=5\Rightarrow h+2k=5$

14. Equation of line L_2 is $\frac{x-3}{1}=\frac{y+4}{-3}=\frac{z-1}{-1}$

(1) equation of plane containing L_1 & $L_2 \qquad \begin{vmatrix} x-3 & y+1 & z-2 \\ 0 & 3 & 1 \\ -1 & 6 & 2 \end{vmatrix}=0\Rightarrow y+3z-5=0$

15. (2) Volume of tetrahedron $=\frac{1}{6}\begin{vmatrix} 3 & -4 & 1 \\ 5 & -4 & 1 \\ 7 & -7 & 0 \end{vmatrix}=\frac{7}{3}$

16. Locus of Q an ellipse whose vertices z1 and z2 don't lie on it and eceentricity is $\sqrt{\frac{2}{3}}$ for locus of Q. Now, $2ae=2\times\sqrt{\frac{2}{3}} \qquad |z_1-z_2|=2\ \therefore[m]=2(\because m<|z_1-z_2|)$

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PHYSICS

18. $\frac{q_1}{r}=\frac{Q'}{R}$ $q_1+Q^1=Q$

$Q\left(\frac{r}{R}+1\right)=Q$ $Q'=\frac{QR}{R+r}, q_1=\frac{Qr}{R+r}$

Similarly $Q''=\frac{QR^2}{(R+r)^2}, q_2=\frac{QrR}{(R+r)^2}$

$$\frac{q_2^3}{q_1}=\frac{\left[\frac{QrR}{(R+r)^2}\right]}{\left(\frac{Qr}{R+r}\right)}=\frac{QR^2r}{(R+r)}=q_3$$

19. The magnitude of friction between the ground and the box is decreasing

20.

$$x_{cm}=\frac{m\left(\frac{R}{2}\right)+2mR}{m+2m}=\frac{5}{6}R$$

$$I_A=I_1+I_2,\ I_1=m\left(\frac{R}{2}\right)^2+m\left(\frac{R}{2}\right)^2=\frac{mR^2}{2}$$

$$I_2=2mR^2+2mR^2=4mR^2\Rightarrow I_A=\frac{9}{2}mR^2$$

For compound pendulum $T=2\pi\sqrt{\frac{1}{Mgd}}$

$\Rightarrow$ here $M=3m,\ d=\frac{5}{6}R\Rightarrow T=2\pi\sqrt{\frac{9R}{5g}}$

21. $\omega=\frac{\pi}{3}, \frac{2\times v\times\sin 45}{9}=1\Rightarrow v=\frac{g\times 1}{2\sin 45}=\frac{10}{\sqrt{2}}=5\sqrt{2}=\sqrt{50}m/s$

22. Balancing torque about contact at ceiling

$4mg\frac{l}{2}\cos\theta+3mg\frac{l}{2}\cos\theta-T_1 l\cos\theta-3mg\frac{l}{2}\cos\theta=0$ $\therefore T_1=2mg$

Also, F_{net} in vertical $=0$ $\therefore F_{hinge}=13mg$

23. $E=6\sin 20t+8\cos 20t$ $=10\sin(20t+\phi_0)$

$\phi_0=\tan^{-1}\left(\frac{8}{6}\right)=53^o$ $E_0=10volt; I_0=2A$

$P=\frac{E_0I_0}{2}10W$ $Q=\omega_0\frac{L}{R}=20\left(\frac{2}{10(5)}\right)=0.8$

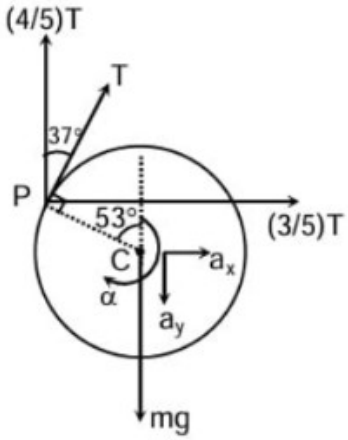
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$$\omega_0 L = \frac{1}{\omega_0 C} \Rightarrow C = \frac{1}{\omega_0^2 L} = 12.5mF$$

24. $\frac{3}{5}T = ma_x \qquad a_x = \frac{3T}{5m} \qquad \ldots\ldots(i)$

$mg - \frac{4}{5}T = ma_y \qquad a_y = g - \frac{4T}{5m} \qquad \ldots\ldots(ii)$

$\tau_c \Rightarrow TR = \frac{1}{2}mR^2\alpha \qquad R\alpha = \frac{2T}{m} \qquad \ldots\ldots(iii)$



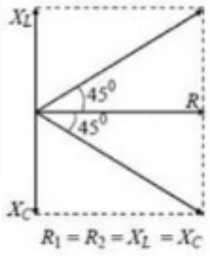
Acceleration of point P along the string 'S_1' is zero

25. Using Bernoulli's equation and continuity equation $l' = \frac{A}{a}\sqrt{\frac{\rho A}{4\mu}}l$

Where H is length of can $l' = 5m$

26. Peak current through $R_1, I_1 = \frac{50}{10\sqrt{2}} = \frac{5}{\sqrt{2}}A$

Peak current through $R_2 \qquad I_2 = \frac{50}{10\sqrt{2}} = \frac{5}{\sqrt{2}}A$



Phase difference between I_1 and I_2 is $\frac{\pi}{2}$

$\therefore$ Peak current through the source is

$$I = \sqrt{I_1^2 + I_2^2} = \sqrt{\left(\frac{5}{\sqrt{2}}\right)^2 + \left(\frac{5}{\sqrt{2}}\right)^2} = \frac{5}{\sqrt{2}} \times \sqrt{2} \quad \therefore I = 5A$$

27. $\left(P_0 + \frac{4S}{R}\right)\left(\frac{4}{3}\pi R^3\right) = \left(\frac{P_0}{2} + \frac{4S \times 4}{5R}\right) \times \frac{4}{3}\pi\left(\frac{5R}{4}\right)^3$

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$$\Rightarrow P_0+\frac{4S}{R}=\frac{125P_0}{128}+\frac{25S}{4R}\Rightarrow P_0=\frac{96S}{R}=\frac{16nS}{R}\therefore n=6$$

28. Speed of wave in the lighter string = (50)(1) = 50 cm/s = $\sqrt{\frac{T}{\mu}}$

Speed of wave in the heavier string $=\sqrt{\frac{T}{4\mu}}=25\,cm/s$

So wavelength in heavier string = 0.5 cm

$\because\Delta\phi=\frac{2\pi}{(1cm)}(9.5cm)=(9.5)(2\pi)$

A point that will oscillation in phase with S must be ahead in phase by (0.5) (2π) with respect to Q. If x is the distance of this point from Q then.

$\frac{2\pi}{0.5cm}x=(0.5)(2\pi)\quad\therefore x=0.25cm$

So, time required is $\frac{9.5}{50}+\frac{0.25}{25}=0.2\sec$

29. $d\phi=\frac{\mu_0 i}{2\pi x}\times 1dx\qquad\phi=\frac{\mu_0 i}{2\pi}1\ln\left(\frac{d_2}{d_1}\right)$

$$\varepsilon=\frac{d\phi}{dt}=\frac{\mu_0}{2\pi}1\ln\left(\frac{d_2}{d_1}\right)\times i_0\omega\cos wt$$

30. $T=\frac{3}{8}M\omega^2L^2$

$$\text{Stress}=\frac{F_\perp}{(A/\cos\theta)}=\frac{3}{8}\frac{M\omega^2L^2\cos^2 q}{A}$$

31. The angular velocity of the rod just after collision is 20 rad/s

32. The angular velocity of the ball just after collision is 10rad/s

33. Position of the point of impact with the horizontal X-Z plane after collision with the wall is (x, y, z). (x + y + z) /100 is 19

34. The ball hits the wall at time t (in sec) when its velocity is making an angle θ (in degrees) with the vertical θ/t is 9

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CHEMISTRY

35. 'l' explains fine spectrum
'm_l' explains start effect

36. $Al^{3+} \times 2 \Rightarrow +6$

$O^{2-} \times 4 \Rightarrow -8$

$\frac{x}{3}(+2)+\frac{2x}{3}(+3)=+2 \Rightarrow x=0.75$

37. II, III, IV, V, VI are correct.

38. $Co(en)_3^{3+}$ *has* 2 *isomers*

$Co(gly)_3$ *has* 4 *isomers*

A mixture with more components has greater entropy

$(\Delta s)_{mix} = -nR\left(\sum x_i \ln x_i\right)$

39. A) $NaNO_{3aq} \xrightarrow{electrons} NaNO_3 + H_2 + O_2$ as concentration of $NaNO_3$ increases, λ decreases

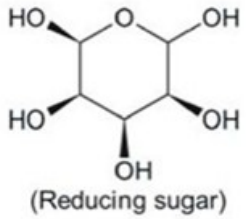
C) $0.1 mol\ AgNO_3 \xrightarrow{0.1F} Ag + O_2 + HNO_3$

$\underset{(0.1 mol)}{HNO_3} \xrightarrow{0.1F} 2H_2 + O_2$

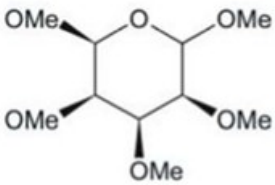
D) $AgNO_3 \xrightarrow{electron's} HNO_3$

$\lambda_{HNO_3} > \lambda_{AgNO_3}$

40. Since anomeric carbon is a chiral carbon, so given sugar can exist in two anomeric pyranose forms. Since it is in hemiacetal, so it can reduce to tollen's reagent to silver mirror.



(Reducing sugar)



41. A) H_3PO_3 in basic with one –'I' group
H_3PO_4 is disbasic with one – I group
B) PF_3 : Show hydrolysis (back bonding) PCl_3F_2 is symmetric

42. Let solubility in pure water = 3

$\therefore K_{SP} = S^2$

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In the salt solution $S' = 1000s$ $\therefore Ag \times \rightleftharpoons AS^+ + x^-$

Initial 0 $\overset{o}{1000}S$ $\overset{o}{1000}S$ $(\alpha = 1)$

$X^- + H^+ \rightleftharpoons H \times 10^8$ $\left[X^-\right]_{eq}\left[AS^+\right] = K_{SP}$

Trace $1000S$ eq^n

$\left(X^-\right)(1000S) = S^2$ and $= \dfrac{(1000S)}{\left(X^-\right)_{eq}(H^+)} = 10^8$ $\therefore \left[H^+\right] = 10^{-2}$

$RNH_3^+ + H_2O \rightleftharpoons RNH_3OH + H^+$ $K_a = \dfrac{10^{-14}}{10^{-12}} = 10^{-2}$

$RNH_3^+ + H_2O \rightleftharpoons RNH_3OH + H^+$

$RNH_3^+ + H_2O \rightleftharpoons RNH_3OH + H^+$

$C(1-n)$ Ch Ch

$\text{Ch} = \left[H^+\right] = 10^{-2} \Rightarrow \dfrac{10^{-4}}{C = Ch} = 10^{-2}$ $C - 10^{-2} = 10^{-2}$

$C = 2 \times 10^{-2} M$ Moles in 500 ml = 10^{-2}

Mass $= 10^{-2} \times 100 = 1\,gm$

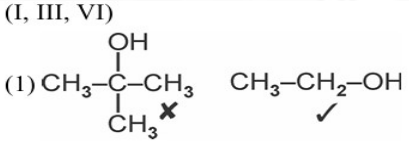
43. $3M + 4H_2O \underset{(g)}{\rightleftharpoons} H_3O_4 \underset{(s)}{+} 3H_2 \underset{(g)}{}$ $R.H = \dfrac{P_{H_2O}}{Aq.tension} \Rightarrow \dfrac{P_{H_2O}}{1} = 0.25$

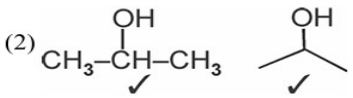
$(\therefore At$ 100°C aq.tension = 1 atm) $P_{H_2} = P_{total} - P_{H_2O(g)}$

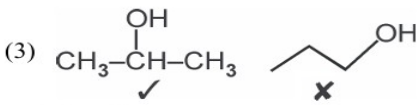
$= 1 - 0.25 = 0.75\,atm = K_P = \dfrac{\left(P_{H_2}\right)^4}{\left(P_{H_2O}\right)} = \dfrac{(3/4)^4}{(1/4)^4} = 81.$

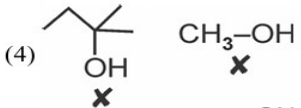
44.

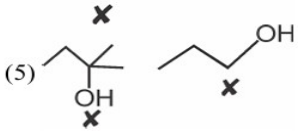
(I, III, VI)

(1) 

(2) 

(3) 

(4) 

(5) 

45. qualitative analysis

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46. $B + 2A^+ = B^{2+} + 2A$

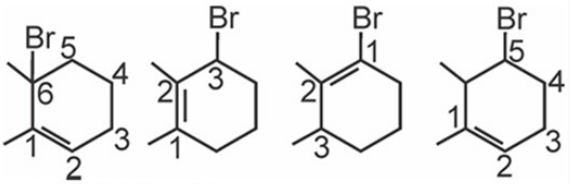
$-1-x \quad \frac{x}{2} \quad x \quad (equilibrium)$

$E^0_{cell} = \frac{0.06}{2}\log K_{eq}$ and x = 0.8M given

$E^0_{A^+/A} - 0.02 = \frac{0.06}{2}\log\frac{(0.4)}{(0.2)^2} = 0.03\text{V} \Rightarrow E^0_{A^+/A} = 0.05V$

$E_{A^+/A} = 0.05 - \frac{0.06}{1}\log\frac{1}{(0.2)} = 0.008\text{ V}$

47.



$6 + 3 + 1 + 5 = 15$

48. As per graph : F.pt of solvent = 278K

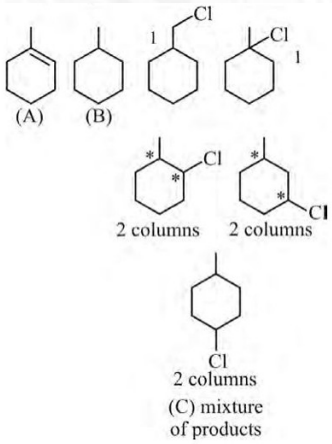
80% dimerised solute $\Rightarrow 1^o = 0.06 \quad \Delta T_f = i K_f m$

$0.6 \times K_f(2) = 7.2 \Rightarrow K_f = \frac{7.2}{0.12} = 6\,Kkg\,mol^{-1}$

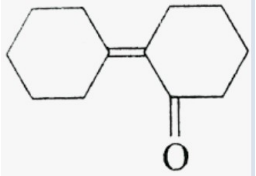
$3 = 6 \times \frac{3}{4} \times m (\because i = 3/4) \Rightarrow m = \frac{2}{3}$

49. $\frac{P_0 - P_s}{P_0} = \frac{2/3}{(1000/100)} = \frac{1}{15}$

50.



51. The product 'E' is



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